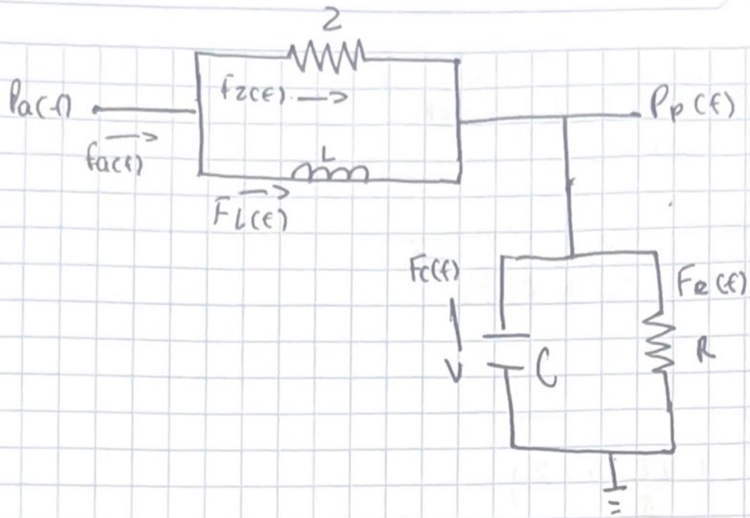

$$Fa(t) = Fz(t) + FL(t) = Fc(t) + FR(t)$$

$$Fz(t) = \frac{Pa(t) - Pp(t)}{z} \qquad FL(t) = \frac{1}{L}\int Pa(t) - Pp(t)\,dt$$

$$Fc(t) = \frac{C\,d(Pp(t)}{dt} \qquad FR = \frac{Pp(t)}{R}$$

$$\frac{Pa(t) - Pp(t)}{z} + \frac{1}{L}\int Pa(t) - Pp(t)\,dt = \frac{C\,dPp(t)}{dt} + \frac{Pp(t)}{R}$$

Laplace: $\frac{Pa(s)}{z} - \frac{Pp(s)}{z} + \frac{Pa(s)}{LS} - \frac{Pp(s)}{LS} = CS + \frac{Pp(s)}{R}$

$$\left(\frac{1}{z} + \frac{1}{LS}\right)Pa(s) = \left(CS + \frac{1}{z} + \frac{1}{LS} + \frac{1}{R}\right)Pp(s)$$

$$\left(\frac{z + LS}{zLS}\right)Pa(s) = \left(\frac{RLCZS^2 + RLS + LSZ + RZ}{RLZS}\right)Pp(s)$$

$$\frac{Pp(s)}{Pa(s)} = \left(\frac{Z + LS}{ZLS}\right)\left(\frac{RLZS}{RLZS\left(CS + \frac{1}{Z} + \frac{1}{R} + \frac{1}{LS}\right)}\right)$$

$$\frac{P_p(s)}{P_a(s)} = \left(\frac{Z+LS}{ZLS}\right)\left(\frac{1}{\frac{ZCRLS^2 + RLS + ZLS + ZR}{ZRLS}}\right)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{RLSZ^2 + ZRL^2S^2}{ZCRL^2Z^2S^3 + RL^2Z^2S^2 + L^2Z^2S^2 + RL^2ZS}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{RLSZ(Z+LS)}{RLSZ\left(CLZS^2 + LS + \frac{LZS}{R} + Z\right)}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{Z+LS}{\frac{CRLZS^2 + LRZ^2S + LZ + ZR}{R}}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{RLS+RZ}{CRLZS^2 + (LRZ^2+LZ)S + ZR} = \frac{RLS+RZ}{CRLZS^2 + (LZ+RL)S + ZR}$$

Modelo de ecuaciones integrodiferenciales

$$P_p(t)\left(\frac{1}{R}+\frac{1}{Z}\right) = \frac{P_a(t)}{Z} + \frac{1}{L}\int\left[P_a(t)-P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L}\int\left[(P_a(t)-P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}\right)\frac{ZR}{Z+R}$$